A HYBRID CNN–SWIN TEMPORAL ATTENTION NETWORK FOR NIGHTTIME VIDEO ANOMALY DETECTION

Yuxuan Jiang[1], *Yunhui Zeng*[1], *Yanlei Cui*[2,3], *and Xin Jin*[1]*

[1]Shenzhen International Graduate School, Tsinghua University, Shenzhen 518000, China
[2]State Key Lab of Intelligent Transportation System, Beihang University, Beijing 100191, China
[3]Hangzhou Innovation Institute, Beihang University, Hangzhou 310056, China
Email: {jyx24, zengyh22}@mails.tsinghua.edu.cn, yanleicui@buaa.edu.cn, jin.xin@sz.tsinghua.edu.cn

ABSTRACT

Video anomaly detection (VAD) in nighttime traffic scenarios remains challenging due to low illumination, overexposure, motion blur, and reflection noise. To address these issues, we propose a Hybrid CNN–Swin Temporal Attention Network (HSTA) for unsupervised nighttime video anomaly detection based on future-frame prediction. HSTA integrates convolutional and transformer-based branches to jointly capture local texture details and global contextual dependencies. A Random Temporal Shift (RTS) module enhances temporal robustness, while a Temporal-Aware Channel Attention Block (TACAB) emphasizes motion-sensitive channels for refined spatial-temporal modeling. Experimental results on real nighttime traffic data demonstrate the effectiveness of HSTA, achieving AUC scores of 76.4%, which is 1.5% higher than the state-of-the-art method. Furthermore, additional evaluations on three daytime benchmarks indicate that HSTA exhibits promising generalization and robustness across different lighting conditions.

***Index Terms*—** Anomaly Detection, Hybrid Network, Random Temporal Shift, Temporal-Aware Channel Attention

1. INTRODUCTION

Nighttime traffic anomaly detection is crucial for road safety and efficient traffic management, as many severe incidents such as collisions, sudden braking, and illegal maneuvers frequently occur under low-light conditions. Compared with daytime scenarios, nighttime video anomaly detection (VAD) is significantly more challenging due to insufficient illumination, complex lighting variations from street lamps and vehicle headlights, strong reflections on wet roads, and motion blur from high-speed vehicles. These factors substantially degrade visual quality and obscure critical cues, hindering the reliability of conventional VAD methods. Therefore, there is an urgent need for a more robust nighttime VAD framework capable of addressing these adverse conditions.

*Corresponding author.

Existing VAD approaches still face significant challenges in nighttime scenarios due to low illumination, motion blur, and local glare. CNN-based methods [1–5] mainly rely on local feature extraction and thus struggle with spatial degradation in extremely dark or overexposed regions. Transformer-based models [6–8] capture long-range dependencies but are sensitive to illumination variations and local noise, often leading to temporal inconsistency. Optical-flow-based techniques [9, 10] exploit motion cues but suffer from unreliable flow estimation under reflective and blurry conditions.

Recent hybrid architectures [11, 12] integrate multiple backbones for richer representations, yet mostly rely on static feature fusion and lack effective spatial–temporal interaction for robust nighttime detection. Meanwhile, LVLM-based or LLM-inspired approaches [13] show strong semantic understanding and cross-scene generalization, but their high computational cost and limited temporal localization hinder real-time nighttime traffic surveillance.

To overcome the aforementioned challenges, we propose a Hybrid CNN–Swin Temporal Attention Network (HSTA) for nighttime traffic anomaly detection. HSTA employs a hybrid encoder that fuses multi-level features from ResNet-50 [14] and the Swin Transformer [15] via cross-level feature concatenation, followed by two parallel branches for spatial and temporal modeling. In the temporal branch, a Random Temporal Shift (RTS) module performs stochastic temporal shifts with varying directions and magnitudes to enhance adaptability to irregular motion and robustness under challenging nighttime conditions. Meanwhile, a decoder equipped with Temporal-Aware Channel Attention Blocks (TACAB) integrates residual channel attention with 3D temporal modeling to refine motion-aware features and improve temporal consistency. Benefiting from these designs, HSTA achieves 76.4% AUC on real nighttime traffic data with an inference speed of 11.05 fps, outperforming state-of-the-art methods while maintaining strong robustness under low-light, reflective, and high-speed motion conditions.

The rest of this paper is organized as follows. Section II details the proposed method. Section III presents comprehen-

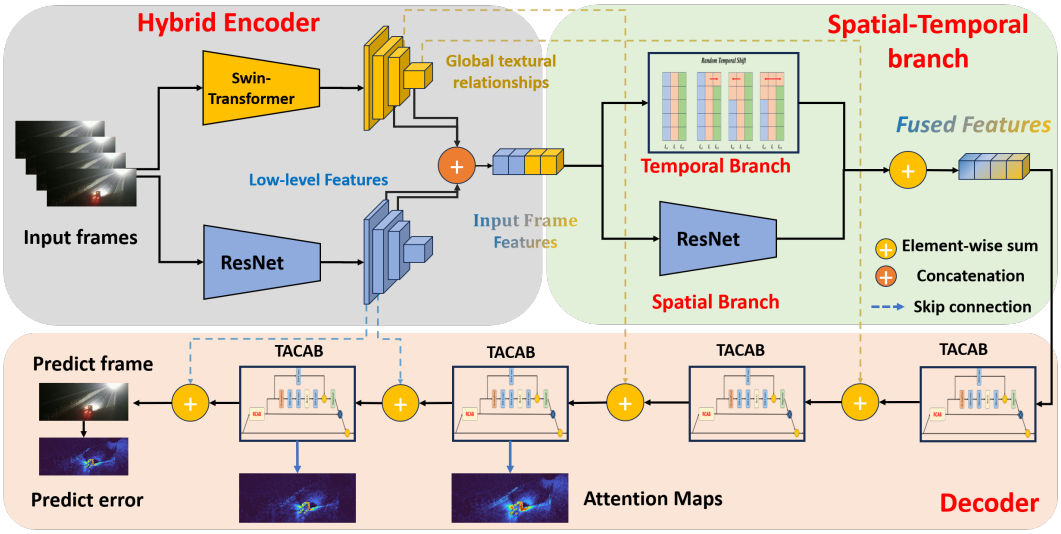


Fig. 1: Overview of the proposed HSTA framework, including the dual-branch encoder, spatial–temporal refinement branch, and TACAB-based decoder.

sive experiments to evaluate the proposed method. Section IV provides a summary of the paper.

2. METHOD

As shown in Fig. 1, the proposed HSTA follows an encoder–decoder architecture that aims to maintain spatial discriminability and temporal consistency in the presence of severe nighttime degradations, facilitating robust video anomaly detection. The framework consists of three tightly coupled components: **(i)** a **hybrid encoder via cross-level feature fusion** that extracts complementary local and global representations from consecutive frames, **(ii)** a **dual-branch spatial-temporal feature refinement** that explicitly models appearance structures and temporal dynamics, and **(iii)** a **temporal attention–guided decoder** equipped with **Temporal-Aware Channel Attention Blocks (TACAB)** for motion-sensitive future-frame reconstruction.

2.1. Hybrid Encoder via Cross-level Feature Fusion

In nighttime surveillance scenarios, a single feature extractor often fails to simultaneously preserve fine-grained local details in low-illumination regions and maintain reliable global contextual awareness under illumination-induced appearance variations. To overcome this limitation, we design a dual-branch hybrid encoder that jointly leverages the complementary strengths of CNNs and Transformers.

Low-level feature branch In the proposed method, we use ResNet-50 to extract detailed appearance cues and localized motion patterns. This branch is sensitive to subtle motion variations, such as human gait or nearby object movement.

Global feature branch In parallel, a Swin Transformer captures hierarchical, long-range dependencies across frames, modeling global context between objects and distant scene elements. This branch is particularly important for understanding interactions between pedestrians or between a pedestrian and a distant object.

To fully exploit complementary representations, we propose a **cross-level feature fusion strategy**. Specifically, we concatenate multi-level features from ResNet and Swin along the channel dimension and apply a 1×1 convolution for projection:

$$F_t = \mathrm{Conv}_{1\times1}\big(\mathrm{Concat}(F^{(l)}_{\mathrm{CNN}}, F^{(l)}_{\mathrm{Swin}})\big), \tag{1}$$

where l denotes the selected feature levels. The fused representation F_t integrates fine-grained spatial details with high-level semantic context for subsequent temporal modeling.

2.2. Dual-Branch Spatial-temporal Feature Refinement

The fused representation $X \in \mathbb{R}^{C\times T\times H\times W}$ of the input sequence frames $\mathcal{I} = \{I_1, I_2, \ldots, I_T\}$ is further processed by two complementary pathways: a temporal branch and a spatial branch, enabling more effective modeling of dynamic and structural information, where $I_t \in \mathbb{R}^{H\times W\times 3}$.

Temporal branch Inspired by the temporal shift strategy widely used in video understanding [16], we introduce a Random Temporal Shift (RTS) module to exploit frame-wise temporal dependencies for anomaly detection, which is illustrated in Fig. 2. Unlike standard temporal shift methods, the proposed RTS introduces stochasticity in both shift direction and magnitude, splitting the channels into groups and shifting each channel along the temporal dimension:

$$\tilde{X}_c(t) = \begin{cases} X_c(t - \Delta t_c), & \text{if left} \\ X_c(t + \Delta t_c), & \text{if right} \\ X_c(t), & \text{otherwise} \end{cases}, \quad \forall c \in [1, C], \quad (2)$$

where $\Delta t_c = \left\lfloor \frac{T}{d_c} \right\rfloor$ denotes the integer temporal offset for channel c, and d_c is randomly sampled from a uniform range.

Spatial branch In parallel, we use ResNet-50 to regularize channel-wise feature distributions, reducing spurious responses caused by illumination variations while maintaining spatial structural integrity for robust anomaly localization.

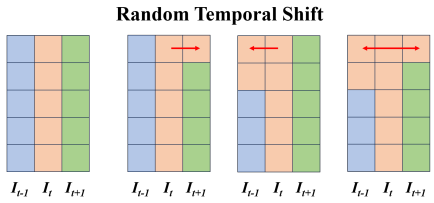


Fig. 2: Illustration of the proposed Random Temporal Shift (RTS) module, which randomly shifts channel-wise features in both direction and magnitude.

2.3. Temporal Attention–Guided Decoder

In VAD tasks, employing a standard convolutional decoder often leads to the loss of temporal information, particularly when anomalous events occur within brief temporal intervals, making precise localization challenging. To address this problem, we design a **Temporal-Aware Channel Attention Decoder** equipped with Temporal-Aware Channel Attention Blocks (TACAB), as shown in Fig. 3, which integrates refined features with multi-scale features from the dual-branch extractor.

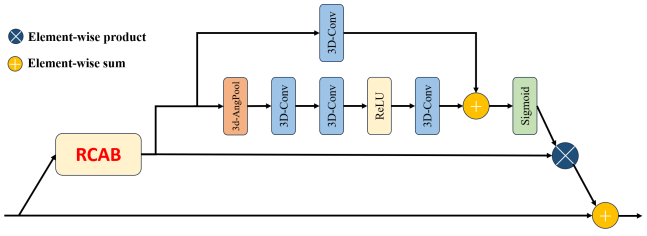


Fig. 3: Illustration of the proposed TACAB.

Formally, given an intermediate feature map F from the spatial–temporal refinement branch, TACAB consists of two parallel pathways, a Residual Channel Attention Block (RCAB) [17] and a temporal-aware branch based on 3D convolutions, where $F \in \mathbb{R}^{C\times T\times H\times W}$.The RCAB refines local spatial features via residual connections and channel attention, which can be formulated as:

$$F_{\text{RCAB}} = F + \mathcal{R}(F), \quad (3)$$

where $\mathcal{R}(\cdot)$ denotes a sequence of convolutional layers followed by channel attention. This branch enhances spatial discriminability while preserving structural consistency.

Temporal-aware branch To explicitly model inter-frame dependencies, we apply a 3D convolution over the spatial and temporal dimensions:

$$F_{\text{3D}} = \mathcal{C}_{3D}(F), \quad (4)$$

where $\mathcal{C}_{3D}(\cdot)$ denotes a 3D convolution operator. This branch captures motion-aware cues and enforces temporal coherence across consecutive frames.

Feature fusion and gating The outputs of the two branches are fused through a channel-wise gating mechanism. Specifically, we compute dynamic channel weights using a sigmoid function:

$$W = \sigma(F_{\text{3D}}), \quad (5)$$

where $\sigma(\cdot)$ is the sigmoid activation. The final output of TACAB is obtained by modulating the RCAB features with these weights:

$$F_{\text{out}} = W \odot F_{\text{RCAB}}, \quad (6)$$

where $\odot$ denotes element-wise multiplication. This gating mechanism adaptively emphasizes motion-relevant channels while suppressing irrelevant responses, thereby improving both spatial discriminability and temporal consistency for video prediction and anomaly detection.

3. EXPERIMENTS

3.1. Experimental Setup

We first conducted the experiments on a real nighttime traffic dataset captured by surveillance cameras equipped on the highway. It consists of 26 highway accident videos with a total duration of 56 minutes, covering various types of traffic incidents, including vehicle collisions, loss of control, road construction, and vehicle explosions. In addition, the videos exhibit challenging visual conditions such as abnormal reflections, overexposure, and motion blur caused by the limitations of surveillance hardware. The dataset is publicly available at: https://github.com/yxJiang0125/Dataset-for-nighttime-traffic-videos.

To further evaluate the generalization capability of the proposed method beyond nighttime scenarios, we also conduct additional experiments on three widely used daytime benchmark datasets, including UCSD Ped2 [18], CUHK Avenue [19], and ShanghaiTech [10], following their standard evaluation protocols.

Both the ResNet and Swin-Transformer backbones are initialized with pretrained weights to leverage prior knowledge and accelerate model convergence. The entire model is trained on 6 GTX 3090 GPUs.

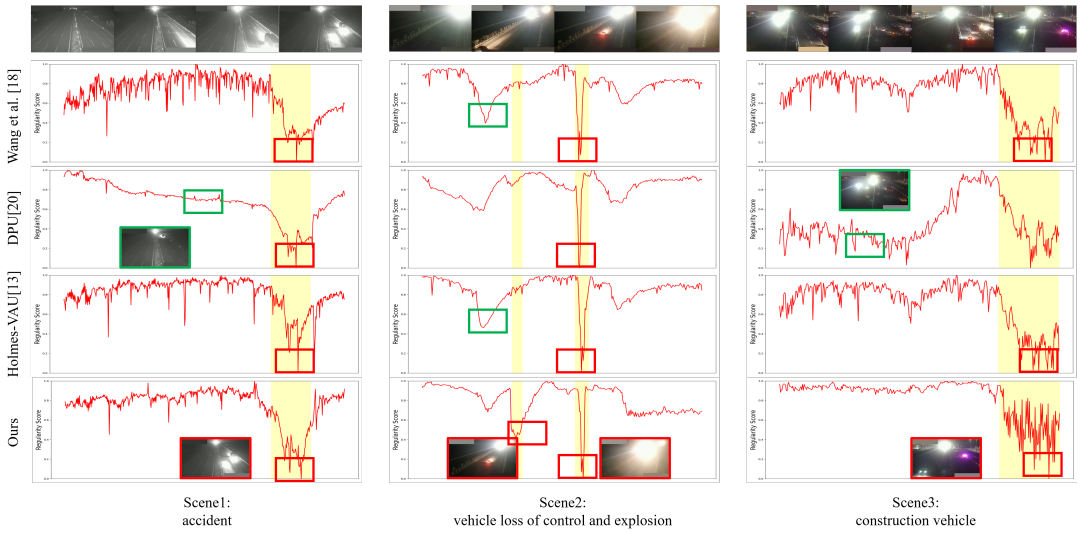


Fig. 4: Qualitative visualization of model performance across different scenes. In each subfigure, red boxes indicate the detected abnormal events, and the corresponding event frames are highlighted with red borders beside the plots. Green boxes denote cases where the model failed to detect an anomaly event or misclassified normal events.

To quantitatively evaluate anomaly detection performance, we adopt the area under the ROC curve (AUC) as the primary metric, which is widely used in video anomaly detection and is independent of specific model architectures.

For intuitive visualization and fair comparison with reconstruction- or prediction-based methods, we compute a frame-wise regularity score $R(t) \in [0,1]$ [10, 20, 21] based on Peak Signal-to-Noise Ratio (PSNR) between the ground-truth frame I_t and the predicted frame $\hat{I}_t$:

$$R(t) = \frac{\text{PSNR}(I_t, \hat{I}_t) - \text{PSNR}_{\min}}{\text{PSNR}_{\max} - \text{PSNR}_{\min}}, \tag{7}$$

where $\text{PSNR}_{\min}$ and $\text{PSNR}_{\max}$ denote the minimum and maximum PSNR values across the video.

For methods that do not explicitly generate future frames [13], we follow the original anomaly scoring strategies (e.g., text- or feature-based anomaly scores) and normalize the resulting frame-wise anomaly scores into $[0,1]$ for consistent AUC computation and visualization. This ensures a fair comparison across heterogeneous models while preserving each method's inherent inference paradigm.

3.2. Results

3.2.1. Results on Nighttime Traffic Videos

To evaluate the effectiveness of the proposed method in realistic and challenging conditions, we compared HSTA with representative generative-based VAD approaches on the nighttime traffic dataset, including CNN–optical flow-based models [10, 21], an attention-based residual autoencoder [20], a hybrid Meta Prototype Network (DPU) [3], and a recent LLM-based method [13]. The evaluation covers typical anomalous events such as vehicle collisions, loss of control, and vehicle fires, as well as challenging visual conditions including reflection noise and overexposure.

Table 1 reports the comparison in terms of AUC and inference speed. HSTA achieves an AUC of 76.4% while processing 11.05 frames per second (fps) under a sliding-window setting with 4 input frames and batch size 1 on a single GTX 3090 GPU, demonstrating a favorable accuracy–efficiency trade-off in complex nighttime environments with severe illumination variations and irregular motion patterns.

Table 1: Performance comparison for different VAD methods on the nighttime traffic dataset.

Method	**AUC** ↑	**Inference speed (fps)** ↑
Chang *et al.* [12]	69.7	10.09
Wang *et al.* [21]	72.7	10.17
ASTNet [20]	72.0	11.22
DPU [3]	69.3	10.93
Holmes-VAU [13]	74.9	9.37
Ours	**76.4**	**11.05**

As shown in Fig. 4, CNN–optical flow-based methods [21] exhibit large performance fluctuations under nighttime conditions due to flickering lights and reflection noise, while purely CNN-based models [20] tend to overfit illumination variations and lose spatial–temporal consistency. The hybrid model [3] also degrades in challenging scenes where strong reflections dominate the Transformer branch and suppress the contribution of CNN features. Although [13]

demonstrates strong long-term anomaly understanding, its transformer-heavy design incurs high computational cost and is susceptible to complex scene interference, which limits temporal localization accuracy.

3.2.2. *Results on Daytime Benchmark Datasets*

As shown in Table 2, HSTA achieves **98.5% on Ped2**, **91.1% on Avenue**, and **74.3% on ShanghaiTech**. These results indicate that HSTA can also generalize effectively to daytime environments with diverse scene layouts, motion patterns, and illumination characteristics. As illustrated in Fig. 5, the proposed method is capable of accurately highlighting when anomalies occur.

Table 2: Performance comparison of different video anomaly detection methods on three benchmark datasets.

Method	Ped2	Avenue	ShanghaiTech
Chang *et al.* [12]	96.5	86.0	73.3
ASTNet [20]	97.4	86.7	73.6
Wang *et al.* [21]	–	90.3	77.8
DPU [3]	95.7	81.7	–
Holmes-VAU [13]	–	–	**78.5**
Ours	**98.5**	**91.1**	74.3

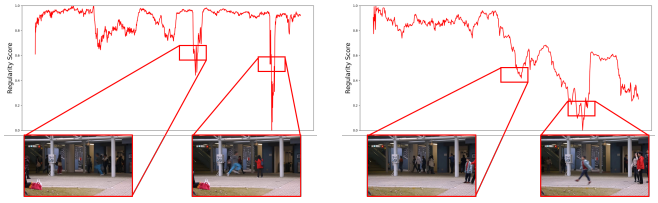


Fig. 5: Visualization of the results for videos in Avenue [19]

3.3. Ablation Study

To validate the contribution of each proposed component, we conducted an ablation study on the nighttime dataset, ShanghaiTech and Avenue. As shown in Table 3, both the Temporal-Aware Channel Attention Block (TACAB) and the Random Temporal Shift (RTS) improve temporal coherence and motion sensitivity.

As illustrated in Fig. 6, attention maps on the nighttime traffic dataset show that TACAB effectively focuses on moving objects and anomalous regions, while RTS enhances inter-frame motion sensitivity. Their joint integration improves both anomaly detection accuracy and interpretability.

4. CONCLUSION

This paper proposes HSTA, a Hybrid CNN–Swin Temporal Attention Network for nighttime video anomaly detection.

Table 3: Ablation study of the proposed modules on three datasets.

Method	Nighttime	Avenue	ShanghaiTech
CNN–Swin only	66.8	86.2	69.5
+RTS	72.9	89.9	73.7
+TACAB	71.6	88.7	71.4
RTS + TACAB	**76.4**	**91.1**	**74.3**

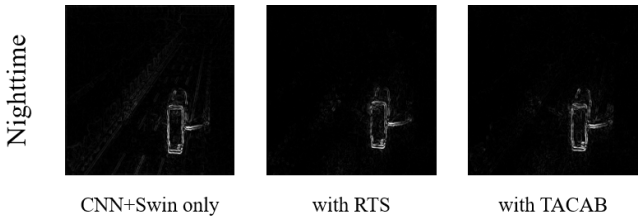


Fig. 6: Visualization of attention maps from ablation experiments.

By integrating CNN-based local feature extraction with Swin Transformer-based global modeling, HSTA captures multi-scale spatio-temporal dependencies. In particular, RTS serves as a temporal regularizer to stabilize modeling under irregular motion, while TACAB performs motion-aware channel refinement to suppress illumination-induced artifacts and reflection noise. These complementary designs jointly enhance motion sensitivity and spatial–temporal consistency in challenging nighttime traffic scenarios.

Experiments on a real nighttime traffic dataset demonstrate that HSTA achieves an AUC of 76.4%, outperforming representative baselines while maintaining competitive inference efficiency (11.05 fps). Furthermore, evaluations on three daytime benchmarks (UCSD Ped2, CUHK Avenue, and ShanghaiTech) validate the cross-domain generalization capability of HSTA, indicating its robustness to diverse scene layouts and lighting conditions.

Acknowledgment

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